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# Data Wrangling Practical - Titanic Dataset
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# Step 1: Import Required Libraries
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import pandas as pd
import numpy as np


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# Step 2: Load Dataset
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# Make sure titanic.csv is uploaded in Google Colab

df = pd.read_csv("titanic.csv")


# -----------------------------------------------
# Step 3: Display First 5 Rows
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print("First 5 Rows of Dataset:")
print(df.head())


# -----------------------------------------------
# Step 4: Dataset Shape and Information
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print("\nShape of Dataset:")
print(df.shape)

print("\nDataset Information:")
print(df.info())


# -----------------------------------------------
# Step 5: Check Missing Values
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```python
print("\nMissing Values:")
print(df.isnull().sum())


# -----------------------------------------------
# Step 6: Statistical Summary
# -----------------------------------------------

print("\nStatistical Summary:")
print(df.describe(include='all'))


# -----------------------------------------------
# Step 7: Check Data Types
# -----------------------------------------------

print("\nData Types of Columns:")
print(df.dtypes)


# -----------------------------------------------
# Step 8: Handle Missing Values
# -----------------------------------------------

# Fill missing Age values using mean
df['Age'] = df['Age'].fillna(df['Age'].mean())

# Fill missing Embarked values using mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Fill missing Cabin values with 'Unknown'
df['Cabin'] = df['Cabin'].fillna('Unknown')

print("\nMissing Values After Handling:")
print(df.isnull().sum())


# -----------------------------------------------
# Step 9: Data Type Conversion
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df['Age'] = df['Age'].astype(float)

df['Survived'] = df['Survived'].astype(int)
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print("\nData Types After Conversion:")
print(df.dtypes)


# ------------------------------------------------
# Step 10: Data Normalization
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# Min-Max Normalization for Age column

df['Age_norm'] = (df['Age'] - df['Age'].min()) / (df['Age'].max() - df['Age'].min())

print("\nNormalized Age Column:")
print(df[['Age', 'Age_norm']].head())


# ------------------------------------------------
# Step 11: Convert Categorical Variables to Numerical
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# Label Encoding for Sex column
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# One Hot Encoding for Embarked column
df = pd.get_dummies(df, columns=['Embarked'])

print("\nDataset After Encoding:")
print(df.head())


# ------------------------------------------------
# Step 12: Final Dataset Information
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print("\nFinal Dataset Shape:")
print(df.shape)

print("\nFinal Dataset Preview:")
print(df.head())
```